

E-COMMERCE FUNNEL ANALYSIS

CONVERSION PERFORMANCE & REVENUE INSIGHTS



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SQL Analysis Overview

Objective

The objective of this SQL analysis is to evaluate e-commerce conversion performance, customer drop-offs, and revenue efficiency using session-level data.

Problem Statement

E-commerce businesses generate large amounts of customer and transaction data, but raw session data alone does not clearly show where potential customers are being lost or where revenue opportunities are being missed.

The key problem is to analyze the customer journey from website visit to completed purchase and identify:

- Where customers drop off during the conversion funnel.
- How different channels and campaigns perform.
- How customer and device segments differ in conversion behaviour.
- How discounts relate to order value and realized revenue.
- How much revenue is lost between expected order value and actual revenue.
- How conversion performance changes over time.

The purpose of this SQL analysis is to transform session-level e-commerce data into measurable business insights that can help identify conversion and revenue optimization opportunities.

Key Analysis Areas

Funnel Conversion

Measured conversion rates and drop-offs across Website → Product → Cart → Checkout → Purchase.

Channel & Campaign Performance

Compared sessions, purchases, conversion rates, revenue, and AOV across channels and campaigns.

Customer & Device Analysis

Analyzed conversion behaviour across user types and devices.

Discount & Revenue Analysis

Evaluated discount usage, order value, realized revenue, and revenue leakage.

Time-Based Analysis

Analyzed monthly conversion rates, purchase volume, and changes in conversion performance.

Q1.What is the overall scale and conversion performance of the e-commerce platform ?

```
SELECT count(SESSION_id) as total_session , count( DISTINCT user_id)
as total_customers
from ecommerce_funnel;
```

```
SELECT COUNT(CASE WHEN purchase_completed = 'yes' THEN 1 END)
AS total_purchases,
COUNT(session_id) AS total_sessions,
COUNT(CASE WHEN purchase_completed = 'yes' THEN 1 END)
/ COUNT(session_id) * 100 AS conversion_rate
FROM ecommerce_funnel;
```

The analysis recorded **120,000 total sessions**, resulting in **8,181 completed purchases** and an overall **conversion rate of 6.82%**.

Q2.Where are customers dropping off across the conversion funnel ?

```
SELECT count(case when viewed_product='yes' then 1 end)
as product_viewed,count(CASE WHEN viewed_product='yes'
and added_to_cart='no' then 1 end) as drop_off_users_after_product_view,
count(CASE WHEN added_to_cart='yes' then 1 end)/
count(CASE WHEN viewed_product='yes' then 1 END)*100 as
conversion_rate from ecommerce_funnel;
```

```
SELECT count(case when added_to_cart='yes' then 1 end)
as cart_users,count(case when checkout_started='yes'
then 1 end) as checkout_users,count(case when added_to_cart
='yes' and checkout_started='no' then 1 end ) as
drop_off_users_after_addedtocart,count(case when
checkout_started='yes' then 1 end)/count(case WHEN
added_to_cart='yes' then 1 end)*100 as conversion_rate
from ecommerce_funnel;
```

```
SELECT count(case when checkout_started='yes'
then 1 end) as checkout_users,count(case WHEN
purchase_completed='yes' then 1 end) as purchased_complete,
count(case when checkout_started='yes' AND
purchase_completed='no' then 1 end) as
drop_off_after_checkout,
count(case when purchase_completed='yes' then 1 end)/
count(case when checkout_started='yes' then 1 end)*100 as conversion_rate
from ecommerce_funnel;
```

Across the conversion funnel:

- **77,870** sessions viewed a product, with a **65.13% drop-off** before adding to cart.
- **27,156** sessions added items to the cart, of which **16,234** proceeded to checkout, giving a **59.72% conversion rate**.
- **16,234** sessions started checkout, with **8,181 purchases completed**, resulting in a **50.39% checkout-to-purchase conversion rate**.

The **largest drop-off occurs between Product View and Add to Cart**, where approximately **65% of product-viewing sessions do not proceed to the cart**. This represents the most significant funnel-stage loss in the analyzed customer journey.

Q3. How does conversion performance vary across acquisition channels ?

```
SELECT channel,
count(session_id) as total_session, count(case when viewed_product='yes'
then 1 end) as viewed_product, count(case WHEN
added_to_cart='yes' then 1 end) as cart_users,
count(case when checkout_started='yes' then 1 end)
as checkout_users, count(case when purchase_completed=
'yes' then 1 end) as purchases, count(case WHEN
purchase_completed='yes' then 1 end)/count(session_id)*100
as conversion_rate, sum(revenue) as total_revenue
from ecommerce_funnel GROUP BY channel ;
```

Channel-level analysis shows that **Email recorded the highest conversion rate at 7.31%**, followed by Social at **6.81%**, Organic at **6.81%**, and Paid Ads at **6.72%**.

Paid Ads generated the highest revenue of ₹7.54M, while Email generated **₹1.81M** from a much smaller number of sessions.

The results show that **traffic volume and conversion efficiency are different metrics**. Paid Ads generated the largest revenue mainly through higher session volume, while Email achieved a higher conversion rate with fewer sessions.

Q4. How does conversion performance vary across user types, and campaign types?

```
SELECT user_type, count(session_id) as total_session,
count(case when purchase_completed='yes' then 1 end)
as purchases, sum(revenue)/count(case WHEN
purchase_completed='yes' then 1 end) as AOV,
sum(revenue) as total_revenue, count(case when purchase_completed=
'yes' then 1 end)/count(session_id)*100 as conversion_rate
from ecommerce_funnel GROUP BY user_type;
```

```
SELECT campaign_type, count(session_id) as total_session,
count(case when purchase_completed='yes' then 1 end)
as purchases, sum(revenue)/count(case WHEN
purchase_completed='yes' then 1 end) as AOV,
sum(revenue) as total_revenue, count(case when purchase_completed=
'yes' then 1 end)/count(session_id)*100 as conversion_rate
from ecommerce_funnel GROUP BY campaign_type;
```

User Type

- New: **6.92% conversion**, AOV ₹2,079.62, Revenue ₹11.23M
- Returning: **6.62% conversion**, AOV ₹2,080.78, Revenue ₹5.79M

Campaign Type

- New Launch: **6.79% conversion**, AOV ₹2,092.61, Revenue ₹5.12M
- Influencer: **6.83% conversion**, AOV ₹2,070.92, Revenue ₹3.37M
- Discount: **6.83% conversion**, AOV ₹2,076.11, Revenue ₹8.53M

Q5. How does conversion performance vary across different channel and campaign combinations?

```
SELECT channel,campaign_type,count(session_id) as total_session,
count(case when purchase_completed='yes' then 1 end)
as purchases,sum(revenue)/count(case WHEN
purchase_completed='yes' then 1 end) as AOV,
sum(revenue) as total_revenue,count(case when purchase_completed=
'yes' then 1 end)/count(session_id)*100 as conversion_rate
from ecommerce_funnel GROUP BY channel,campaign_type;
```

Result

- **Email + Discount** recorded the highest conversion rate at **7.43%**.
- **Email + New Launch** achieved a conversion rate of **7.17%**.
- **Organic + Discount** recorded a **7.06%** conversion rate.
- **Paid Ads + Influencer** achieved **6.98%** conversion.
- **Social + Influencer** recorded the lowest conversion rate at **6.48%**.
- **Paid Ads + Discount** generated the highest revenue at approximately **₹3.68M**.

Business Insight

- Conversion performance varies across different **channel–campaign combinations**.
- **Discount campaigns** show relatively strong conversion across several channels.
- **Paid Ads + Discount** generated the highest revenue, mainly supported by its larger session volume.
- This indicates that campaign performance should be evaluated together with the **acquisition channel**, rather than looking at either factor independently.

Q6. How does discount usage relate to purchase behaviour and order value ?

```
SELECT discount_applied, count(session_id) as total_session,
count(case when purchase_completed='yes' then 1 end)
as purchases, sum(revenue)/count(case WHEN
purchase_completed='yes' then 1 end) as AOV,
sum(revenue) as total_revenue, count(case when purchase_completed=
'yes' then 1 end)/count(session_id)*100 as conversion_rate
from ecommerce_funnel GROUP BY discount_applied;
```

Result

- Non-discount sessions: **115,538 sessions**, **3,719 purchases**, **3.22% conversion rate**.
- Discount sessions: **4,462 sessions**, **4,462 purchases**, **100% conversion rate**.
- AOV: **₹2,200.98** without discount vs **₹1,979.20** with discount.

Business Insight

- Discount-applied sessions show **higher observed conversion but lower AOV**.
- The 100% discount conversion rate is a **dataset characteristic**, so it should not be interpreted as a causal effect of discounts.

Q7. How much revenue is lost between order value and actual realized revenue ?

```
SELECT count(case when purchase_completed='yes' THEN
1 end) as purchases, sum(case when purchase_completed='yes'
then order_value else 0 end) as total_OV,
sum(case when purchase_completed='yes' then
revenue else 0 end) as total_revenue,
sum(case when purchase_completed='yes' then order_value
else 0 end)-sum(case when purchase_completed='yes' THEN
revenue else 0 end)
as revenue_leakage, (sum(case when purchase_completed='yes' then order_value
else 0 end)-sum(case when purchase_completed='yes' THEN
revenue else 0 end))/sum(case when purchase_completed='yes'
then order_value else 0 end)*100 as leakage_percent
from ecommerce_funnel;
```

- **Total Purchases:** 8,181
- **Total Order Value:** ₹17.998M
- **Actual Revenue:** ₹17.017M

- **Revenue Leakage:** ₹981,238
- **Leakage Percentage:** 5.45%

₹981K of potential order value was not realized as actual revenue, resulting in a 5.45% revenue leakage.

Q8. How does conversion performance and revenue efficiency vary over time and across channels ?

```
with monthly as (SELECT month,count(session_id)
as total_session,count(case when purchase_completed='yes' then 1
end) as purchases,count(case when purchase_completed='yes' then
1 end )/count(session_id)*100 as conversion_rate
from ecommerce_funnel GROUP BY month)
SELECT month,total_session,purchases,conversion_rate,
LAG(conversion_rate) over (order by month) as
previous_cr,conversion_rate-LAG(conversion_rate)
over (order by month) as CR_change
from monthly ORDER BY month;
```

- **September 2025** recorded the highest conversion rate at **7.06%**.
- Conversion declined to **6.61%** in **November 2025**.
- The rate recovered slightly to **6.82%** in **December 2025**.
- Month-over-month changes show **moderate fluctuations** in conversion performance.
- Conversion performance varied across the analyzed months rather than following a consistent upward or downward trend.
- The decline after September suggests a period that may require further investigation into changes in traffic, campaigns, or customer behaviour.

Q9. Which acquisition channels generate the highest revenue efficiency per 1,000 sessions ?

```
SELECT CHANNEL,count(session_id) as total_session,  
count(case when purchase_completed='yes' then 1 end)  
as purchases,sum(revenue) as total_revenue,count(  
case when purchase_completed='yes' then 1 end)/  
count(session_id)*100 as CR,sum(revenue)/count(session_id)  
*1000 as revenue_per_1000_session FROM ecommerce_funnel  
GROUP BY channel;
```

- Email generated the highest revenue per 1,000 sessions at ₹149,793.
 - Social generated approximately ₹142,684 per 1,000 sessions.
 - Organic generated approximately ₹141,621 per 1,000 sessions.
 - Paid Ads generated approximately ₹139,841 per 1,000 sessions.
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- Email traffic shows the highest revenue efficiency, despite having the lowest session volume.
 - Paid Ads generated the highest total revenue, but its revenue efficiency per 1,000 sessions was the lowest among the four channels.
 - This highlights the difference between traffic volume and revenue efficiency.

Q10. How does discount usage vary between new and returning customers ?

```
SELECT user_type, device, discount_applied,
count(session_id) as total_session,
count(case when purchase_completed='yes' then 1 end) AS
purchases, count(case when purchase_completed='yes' then 1 end)/
count(session_id)*100 as conversion_rate, sum(revenue) AS
total_revenue, AVG(order_value) as AOV from ecommerce_funnel
GROUP BY user_type, device, discount_applied;
```

- **New + Mobile + Discount:** 2,066 sessions, **100% conversion**
 - **New + Desktop + Discount:** 893 sessions, **100% conversion**
 - **Returning + Mobile + Discount:** 1,056 sessions, **100% conversion**
 - **Returning + Desktop + Discount:** 447 sessions, **100% conversion**
 - Without discounts, conversion rates across the user/device combinations remained around **3.10%–3.32%**.
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- Discount-applied sessions show **100% observed conversion across both user types and devices**.
 - Without discounts, conversion remained close to **3%**, with only small differences between devices and user types.
 - The 100% conversion pattern reflects the **dataset structure**, so it should not be interpreted as proof that discounts caused purchases.

SQL Analysis Conclusion

The SQL analysis examined the complete customer journey from Website Visit → Product View → Add to Cart → Checkout → Purchase.

Key Findings

- The overall purchase conversion rate was 6.82% across 120,000 sessions.
- 77,870 sessions reached the product-view stage, but the Product → Cart stage had the highest drop-off at 65.13%.
- 27,156 sessions added products to their cart, while 16,234 sessions proceeded to checkout.
- 8,181 purchases were completed from 16,234 checkout sessions, resulting in a 50.39% Checkout → Purchase conversion rate.
- The analysis identified approximately ₹981K in revenue leakage, representing 5.45% of total order value.
- Channel and campaign analysis showed that conversion efficiency and revenue generation can vary across acquisition combinations.
- Monthly analysis showed fluctuations in conversion performance, with September 2025 reaching the highest conversion rate of 7.06%.